

RUSTAQ, Oman

WMO: 412530

Lat: 23.4092N

Lon: 57.4285E

Elev: 322

StdP: 97.52

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.7	13.9	-2.0	3.3	33.4	0.1	3.9	33.5	7.7	24.9	6.0	21.9	1.1	200	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.5	45.0	21.6	44.0	21.8	42.9	21.9	28.4	35.9	27.9	35.3	27.4	34.8	3.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	23.2	32.3	26.1	22.4	32.6	25.5	21.6	32.5	94.5	36.2	92.0	35.2	89.7	35.0	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.3	5.3	4.4	10.3	47.0	1.8	0.7	9.0	47.5	7.9	47.9	6.9	48.3	5.7	48.8		
			7.3	29.5	1.8	0.5	5.9	29.8	4.9	30.0	3.8	30.3	2.5	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.3	19.9	21.9	25.3	30.1	35.1	36.8	36.5	35.4	33.5	30.2	25.0	21.2
	DBStd	6.46	2.01	2.72	2.69	2.53	2.07	2.07	2.09	2.13	1.62	1.88	2.01	1.84
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	15	9	4	0	0	0	0	0	0	0	0	0	2
	CDD10.0	7036	306	332	475	603	778	805	822	788	706	624	450	346
	CDD18.3	4009	56	103	217	353	520	555	564	529	456	366	200	89
	CDH23.3	57918	274	763	2132	4717	8579	9501	9548	8625	6958	4683	1674	464
	CDH26.7	38332	31	237	989	2750	6117	7106	7069	6147	4579	2631	611	67
Wind	WSAvg	1.9	1.7	1.9	2.0	2.0	2.1	2.1	2.0	2.1	1.9	1.8	1.7	1.6
Precipitation	PrecAvg	136	12	28	17	18	8	7	11	6	6	12	10	12
	PrecMax	209	66	115	105	69	38	58	33	16	23	84	37	37
	PrecMin	57	0	12	0	0	0	0	3	0	0	0	0	2
	PrecStd	48	18	26	25	20	10	17	8	5	7	21	11	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	33.7	36.8	41.0	45.6	46.5	45.8	45.0	42.7	40.0	34.0	30.0
		MCWB	17.2	17.8	18.0	19.4	20.9	21.0	22.4	22.3	21.0	19.8	18.7	17.6
	2%	DB	27.5	31.4	35.3	39.3	43.9	45.2	44.5	43.6	41.2	38.5	32.5	28.4
		MCWB	17.5	17.6	18.1	19.2	20.6	21.1	22.8	22.9	21.3	19.6	19.4	17.8
	5%	DB	26.2	29.4	33.6	38.0	42.7	44.1	43.2	42.2	40.2	37.3	31.4	27.2
		MCWB	17.1	17.6	18.0	18.9	20.4	21.1	23.1	22.9	21.7	19.4	19.3	18.0
	10%	DB	24.9	27.7	31.9	36.5	41.5	42.9	41.9	41.0	39.0	36.0	30.2	26.0
		MCWB	16.8	17.4	17.8	18.8	20.3	21.5	23.7	23.4	22.2	19.4	19.3	17.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.2	21.6	23.0	24.1	26.7	28.8	29.2	28.5	27.6	25.9	23.5	21.1
		MCDB	25.2	27.0	28.5	30.3	33.2	36.0	36.7	36.2	34.1	30.8	27.8	24.9
	2%	WB	19.2	20.6	22.0	23.2	25.1	27.9	28.6	28.0	26.9	25.1	22.5	20.3
		MCDB	23.9	25.7	27.6	29.3	32.6	35.1	36.2	35.5	33.5	29.5	27.5	24.5
	5%	WB	18.6	19.8	21.2	22.2	23.7	27.2	28.1	27.5	26.4	24.4	21.7	19.7
		MCDB	23.3	25.0	27.0	29.1	33.9	34.3	35.7	34.9	32.9	29.4	27.3	24.0
	10%	WB	17.9	19.1	20.4	21.3	22.6	26.5	27.7	27.0	25.8	23.6	21.0	19.1
		MCDB	22.7	24.5	26.7	30.0	35.7	34.4	35.3	34.2	32.4	29.2	26.7	23.4
Mean Daily Temperature Range	5% DB	MDBR	9.9	11.1	11.7	11.7	12.4	11.5	10.1	10.8	11.2	11.6	10.2	9.5
		MCDBR	11.4	13.1	13.8	13.3	13.4	12.9	11.7	12.0	12.4	12.9	11.1	10.9
		MCWBR	3.9	4.4	4.4	3.5	3.5	3.6	4.0	3.7	3.9	4.3	3.7	3.6
	5% WB	MCDBR	9.7	10.2	10.3	10.5	11.6	10.2	9.5	10.0	10.0	10.2	9.7	8.9
		MCWBR	4.0	4.1	4.1	4.2	5.0	4.3	4.1	3.7	3.8	4.1	3.5	3.2
Clear-Sky Solar Irradiance	taub	0.463	0.527	0.610	0.678	0.706	0.778	0.842	0.808	0.695	0.563	0.496	0.463	
	taud	2.015	1.847	1.664	1.571	1.528	1.444	1.391	1.426	1.568	1.788	1.948	2.011	
	Ebn at Noon	783	758	717	680	658	608	570	588	645	716	744	764	
	Edh at Noon	157	199	249	278	289	312	329	318	271	208	167	153	
All-Sky Solar Radiation	RadAvg	4.62	5.42	6.28	6.92	7.55	7.47	6.95	6.79	6.39	5.81	4.83	4.37	
	RadStd	0.23	0.26	0.35	0.42	0.19	0.26	0.26	0.17	0.13	0.18	0.14	0.22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	+0.42	+0.68	N/A	N/A	+0.28	+0.30	N/A	N/A	+96	+98

Nomenclature: See separate page